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RESEARCH ARTICLE

# Hyperbolicity of solution semigroups for linear neutral differential equations

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**Abstract** Consider the linear neutral functional differential equation of the form

$$\begin{cases} \frac{\partial}{\partial t} F u_t = B F u_t + \Phi u_t, & t \geq 0, \\ u_0(s) = \varphi(s), & s \in [-r, 0], \end{cases}$$

where the function $u(\cdot)$ takes values in a Banach space $X$. Under appropriate conditions on the difference operator $F$ and the delay operator $\Phi$ we prove that the solution semigroup for this equation is hyperbolic provided that the differential operator $B$ generates a hyperbolic semigroup on $X$.

**Keywords** Neutral functional differential equations · Semigroups · Existence and uniqueness of solutions · Hyperbolicity of solution semigroups

## 1 Introduction

In this paper we study the hyperbolicity (or exponential dichotomy) of solutions to linear neutral differential equations of the form

$$\begin{cases} \frac{\partial}{\partial t} F u_t = B F u_t + \Phi u_t, & \text{for } t \geq 0, \\ u_0(s) = \varphi(s), & \text{for } s \in [-r, 0]. \end{cases} \tag{1.1}$$

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Here in most applications $B$ is some linear partial differential operator, while the operators $F$ and $\Phi$ are called *difference operator* and *delay operator*, respectively. Also, the corresponding *history function* is defined as

$$u_t(s) := u(t+s) \quad \text{for all } t \geq 0,\ s \in [-r, 0].$$

For the well-posedness of these equations we refer to Hale [5, 6], Wu [15, Chap. 2.3], Wu and Xia [16], Adimy and Ezzinbi [1], Nagel and Huy [11]. Especially, in [11] we have proposed an abstract treatment of these equations by choosing a Banach space $X$ and considering the solution $u(\cdot)$ as a function from $[-r, \infty)$ to $X$ for some positive constant $r$.

Moreover, $B$ is a linear operator on $X$ (representing the partial differential operator), while $F$ and $\Phi$ are linear operators from an $X$-valued function space, e.g., $C([-r, 0], X)$ into $X$. More precisely, we make the following assumption throughout the paper.

**Assumption 1.1** On the Banach spaces $X$ and $E := C([-r, 0], X)$ we consider the following operators.

(i) Let $(B, D(B))$ be the generator of a strongly continuous semigroup $(e^{tB})_{t\geq 0}$ on $X$ satisfying $\|e^{tB}\| \leq Me^{\omega_1 t}$ for some constants $M \geq 1$ and $\omega_1 \in \mathbb{R}$.
(ii) Let the *difference operator* $F : E \to X$ and the *delay operator* $\Phi : E \to X$ be bounded and linear.

In [11], under these assumptions we have solved (1.1) by constructing an appropriate strongly continuous semigroup on the space $E$. This semigroup was obtained by proving that a certain operator (see Definition 2.3) satisfies the Hille-Yosida conditions as long as we can write the difference operator as $F = \delta_0 - \Psi$ with $\Psi$ being "small" (see (2.5)). It was also shown that the smallness of $\Psi$ can be replaced by the condition "having no mass in 0" (see Definition 2.5).

In the present paper, we will prove that under the Assumption 1.1, if the semigroup $(e^{tB})_{t\geq 0}$ is hyperbolic and the norm of the delay operator $\Phi$ is sufficiently small then the semigroup solving (1.1) is also hyperbolic. Our results extend the results known for delay functional differential equations (see Engel [2], Huy [8], Hale and Verduyn Lunel [7], Wu [15]).

## 2 Neutral semigroups

In this section, we briefly recall the results obtained in [11] on the well-posedness of (1.1) as well as the representation of the resolvent of the semigroup solving (1.1). Under the assumptions from Sect. 1, we consider the Banach space $E := C([-r, 0], X)$ endowed with the sup-norm and the operator $(G_m, D(G_m))$ on $E$, defined by

$$\begin{aligned} D(G_m) &:= C^1([-r, 0], X), \\ G_m f &:= f' \quad \text{for } f \in D(G_m). \end{aligned} \tag{2.1}$$

We are now looking at various restrictions of this (maximal) operator yielding generators of strongly continuous semigroups. We start with a simple case.

**Definition 2.1** On the space $E$ we define the operator $T_{B,0}(t)$ by

$$[T_{B,0}(t)f](s) = \begin{cases} f(s+t), & s+t \le 0, \\ e^{(t+s)B} f(0), & s+t \ge 0, \end{cases}$$

for $f \in E$, $t \ge 0$, and $s \in [-r, 0]$.

Moreover, we define the operator $(G_{B,0}, D(G_{B,0}))$ by

$$\begin{aligned} D(G_{B,0}) &:= \{f \in D(G_m) : f(0) \in D(B) \text{ and } f'(0) = Bf(0)\}, \\ G_{B,0} f &:= f' \quad \text{for } f \in D(G_{B,0}). \end{aligned}$$

We then have the following properties of $G_{B,0}$ and $(T_{B,0}(t))_{t\ge 0}$ which have been proved in [8, Proposition 2.8] (see also [3, Proposition VI.6.1]).

**Proposition 2.2** *The following assertions hold.*

(i) $(T_{B,0}(t))_{t\ge 0}$ *is a strongly continuous semigroup on the space* $E$ *with the generator* $(G_{B,0}, D(G_{B,0}))$.

(ii) *The inclusion* $\rho(B) \subset \rho(G_{B,0})$ *holds, and for* $\lambda \in \rho(B)$ *the resolvent* $R(\lambda, G_{B,0})$ *is given by*

$$[R(\lambda, G_{B,0})f](s) = e^{\lambda s} R(\lambda, B) f(0) + \int_s^0 e^{\lambda(s-\xi)} f(\xi) d\xi$$
$$\text{for } f \in E, \ s \in [-r, 0]. \tag{2.2}$$

(iii) *The semigroup* $(T_{B,0}(t))_{t\ge 0}$ *satisfies*

$$\|T_{B,0}(t)\| \le M e^{\omega_1 t}, \quad t \ge 0, \tag{2.3}$$

*with the constants* $M$ *and* $\omega_1$ *appearing in Assumption* 1.1.

We then take the delay operator $\Phi$ and the difference operator $F$ (see (2.5) and Theorem 2.4) to define a different restriction of the operator $G_m$.

**Definition 2.3** The operator $G_{B,F,\Phi}$ is defined by

$$\begin{aligned} G_{B,F,\Phi} f &:= f' \quad \text{on the domain,} \\ D(G_{B,F,\Phi}) &:= \{f \in D(G_m) : Ff \in D(B) \text{ and } F(f') = BFf + \Phi f\}. \end{aligned} \tag{2.4}$$

In [11] we have found conditions on $F$ such that the operator $G_{B,F,\Phi}$ becomes the generator of a strongly continuous semigroup. To do so, we wrote $F$ in the form

$$Ff := f(0) - \Psi f, \quad f \in E, \tag{2.5}$$

for some bounded linear operator $\Psi : E \to X$. The domain of $G_{B,F,\Phi}$ can then be rewritten as

$$D(G_{B,F,\Phi}) = \{f \in D(G_m) : f(0) - \Psi f \in D(B) \text{ and } f'(0) = B(f(0) - \Psi f) + \Phi f + \Psi f'\}.$$

We now recall the theorem on well-posedness of (1.1). It says that if the operator $\Psi$ is "small", then $G_{B,F,\Phi}$ generates a strongly continuous semigroup solving (1.1).

To that purpose and for $\lambda \in \mathbb{C}$ we define the operator $e_\lambda : X \to E$ by

$$[e_\lambda x](s) := e^{\lambda s} x \quad \text{for } s \in [-r, 0],\ x \in X. \tag{2.6}$$

**Theorem 2.4** [11, Theorem 2.4, Corollaries 2.5, 2.6] *Assume that the difference operator $F$ is of the form (2.5) such that $\Psi$ satisfies the condition $\|\Psi\| < 1$. Then the following assertions hold.*

(i) *$\lambda \in \rho(G_{B,F,\Phi})$ for each $\lambda > \omega_1 + \frac{M\|\Phi\|}{1-\|\Psi\|}$ (with the constants $\omega_1$ and $M$ as in Assumption 1.1). For such $\lambda$ the resolvent of $G_{B,F,\Phi}$ has the form*

$$R(\lambda, G_{B,F,\Phi})f = e_\lambda[\Psi R(\lambda, G_{B,F,\Phi}) + R(\lambda, B)(\Phi R(\lambda, G_{B,F,\Phi}) - \Psi)]f + R(\lambda, G_{B,0})f \quad \text{for } f \in E. \tag{2.7}$$

(ii) *The operator $G_{B,F,\Phi}$ generates a strongly continuous semigroup $(T_{B,F,\Phi}(t))_{t\geq 0}$ on $E$.*

(iii) *Equation (1.1) is well-posed. More precisely, for every $\varphi \in D(G_{B,F,\Phi})$ there exists a unique classical solution $u_t(\cdot, \varphi)$ of (1.1) given by*

$$u_t(\cdot, \varphi) = T_{B,F,\Phi}(t)\varphi,$$

*and for every sequence $(\varphi_n)_{n\in\mathbb{N}} \subset D(G_{B,F,\Phi})$ satisfying $\lim_{n\to\infty} \varphi_n = 0$, one has*

$$\lim_{n\to\infty} u_t(\cdot, \varphi_n) = 0$$

*uniformly in compact $t$-intervals.*

We would like to note that, instead of assuming $\Psi$ to be small, we may suppose that $\Psi$ *has no mass in* 0 (see Definition 2.5 below). The idea is to renorm the space $C([-r, 0], X)$ such that, with the new equivalent norm, the norm of $\Psi$ is small. This idea has been used, e.g., by Schwarz [14] to study ordinary neutral functional differential equations via dissipativity conditions. We now recall the notion of "no mass in 0".

**Definition 2.5** A bounded linear operator $\Psi \in \mathcal{L}(C([-r, 0], X), X)$ is said to have *no mass in* 0 if for every $\epsilon > 0$ there exists a positive number $\delta \leq r$ such that

$$\|\Psi(f)\|_X \leq \epsilon \|f\|_\infty \quad \text{for all } f \in C([-r, 0], X) \text{ satisfying } \operatorname{supp} f \subseteq [-\delta, 0].$$

*Remark 2.6* This definition is taken from [14, Definition II.2.1]. We note that, if $\Psi \in \mathcal{L}(C([-r, 0], X), X)$ has the form

$$\Psi(f) = \int_{-r}^{0} [d\eta(\theta)] f(\theta)$$

for some function $\eta(\cdot)$ of bounded variation, then the above definition is equivalent to the fact that the function $\eta(\cdot)$ is nonatomic at 0 in the sense of Hale and Verduyn Lunel [7, Chap. 9.2] or Wu [15, Chap. 2.3].

The renorming of the space $E := C([-r, 0], X)$ can be done as in [11]. Indeed, for each positive number $\omega$, the new norm $\|\cdot\|_\omega$ defined by

$$\|f\|_\omega := \sup_{-r\le s\le 0} \|f(s)e^{-\omega s}\|_X, \quad f \in E, \tag{2.8}$$

is equivalent to the sup-norm. Furthermore, we denote by $C_\omega$ the space $C([-r, 0], X)$ endowed with the norm $\|\cdot\|_\omega$. The following Theorem is taken form [11].

**Theorem 2.7** [11, Lemma 3.3, Theorem 3.4, Corollary 3.5] *Let the operator $\Psi \in \mathcal{L}(E, X)$ have no mass in 0 and $F$ be of the form (2.5). Then, the following assertions hold.*

(i) *There exists a positive number $\omega$ such that the norm of the operator $\Psi$, as a bounded linear operator from $C_\omega$ into $X$, is smaller than 1.*
(ii) *The operator $(G_{B,F,\Phi}, D(G_{B,F,\Phi}))$ is the generator of a strongly continuous semigroup $(T_{B,F,\Phi}(t))_{t\ge 0}$ on $C([-r, 0], X)$.*
(iii) *For every $\varphi \in D(G_{B,F,\Phi})$, there exists a unique classical solution $u_t(\cdot, \varphi)$ of (1.1), given by $u_t(\cdot, \varphi) = T_{B,F,\Phi}(t)\varphi$, where the strongly continuous semigroup $(T_{B,F,\Phi}(t))_{t\ge 0}$ is generated by the operator $G_{B,F,\Phi}$. Moreover, for every sequence $(\varphi_n)_{n\in\mathbb{N}} \subset D(G_{B,F,\Phi})$ satisfying $\lim_{n\to\infty} \varphi_n = 0$, one has $\lim_{n\to\infty} u_t(\cdot, \varphi_n) = 0$ uniformly in compact $t$-intervals.*

## 3 Hyperbolicity

Having established the well-posedness of (1.1), we now consider the hyperbolicity of the solution semigroup $(T_{B,F,\Phi}(t))_{t\ge 0}$. To do this, we first compute the spectra of the semigroup $(T_{B,0}(t))_{t\ge 0}$ on $E = C([-r, 0], X)$ and its generator. This will be used to prove the robustness of the hyperbolicity of the semigroup $(T_{B,F,\Phi}(t))_{t\ge 0}$ under small perturbations by the delay operator $\Phi$. We first compare $(T_{B,0}(t))_{t\ge 0}$ to its restriction to the invariant subspace $C_0 := \{f \in E : f(0) = 0\}$.

**Theorem 3.1** *Let the semigroup $(T_{B,0}(t))_{t\ge 0}$ on $E$ be defined as in Definition 2.1. Denote by $(T_0(t))_{t\ge 0}$ the restriction of $(T_{B,0}(t))_{t\ge 0}$ to the subspace $C_0$. Then, the following inclusion holds.*

$$\sigma(T_{B,0}(t)) \subseteq \sigma(T_0(t)) \cup \sigma(e^{tB}) \quad \text{for } t \ge 0. \tag{3.1}$$

*Proof* Endow $X \oplus C_0$ with the 1-norm

$$\|(x, f)\| := \|f\| + \|x\| \quad \text{for } (x, f) \in X \oplus C_0.$$

For a fixed function $\varphi \in C[-r, 0]$ satisfying $\varphi(0) = 1$ we consider the linear operator

$$\mathcal{J} : E = C([-r, 0], X) \to X \oplus C_0, \tag{3.2}$$

$$f \mapsto (f(0), f - \varphi(\cdot) f(0)). \tag{3.3}$$

Then $\mathcal{J}$ is an isomorphism and its inverse is given by

$$\mathcal{J}^{-1} : X \oplus C_0 \to E, \tag{3.4}$$

$$(x, f) \mapsto f + \varphi(\cdot)x. \tag{3.5}$$

Therefore, by similarity, the operators

$$\hat{T}(t) := \mathcal{J} T_{B,0} \mathcal{J}^{-1} = \begin{pmatrix} e^{tB} & 0 \\ (T_{B,0}(t) - e^{tB})\varphi(\cdot) & T_0(t) \end{pmatrix}, \quad t \geq 0,$$

form a semigroup satisfying $\sigma(\hat{T}(t)) = \sigma(T_{B,0}(t))$. Let now $\lambda \in \rho(T_0(t)) \cap \rho(e^{tB})$. Then the operator

$$\begin{pmatrix} \lambda - e^{tB} & 0 \\ (T_{B,0}(t) - e^{tB})\varphi(\cdot) & \lambda - T_0(t) \end{pmatrix}$$

is invertible with the inverse

$$\begin{pmatrix} (\lambda - e^{tB})^{-1} & 0 \\ -(\lambda - T_0(t))^{-1}[(T_{B,0}(t) - e^{tB})\varphi(\cdot)](\lambda - e^{tB})^{-1} & (\lambda - T_0(t))^{-1} \end{pmatrix}.$$

Hence $\lambda \in \rho(\hat{T}(t)) = \rho(T_{B,0}(t))$. This means that $\rho(T_0(t)) \cap \rho(e^{tB}) \subseteq \rho(T_{B,0}(t))$.

Thus, Inclusion (3.1) follows. □

Using this theorem and the spectral properties of nilpotent semigroups, we obtain the following corollary describing the hyperbolicity of the semigroup $(T_{B,0}(t))_{t\geq 0}$.

**Corollary 3.2** *If the operator $(B, D(B))$ generates a hyperbolic semigroup $(e^{tB})_{t\geq 0}$ then the semigroup $(T_{B,0}(t))_{t\geq 0}$ is hyperbolic.*

*Proof* By the definition of $(T_0(t))_{t\geq 0}$ as the restriction of $(T_{B,0}(t))_{t\geq 0}$ to the subspace $C_0$ we have

$$[T_0(t) f](s) = \begin{cases} f(s + t), & s + t \leq 0, \\ 0, & s + t \geq 0, \end{cases}$$

for $f \in C_0$ and $t \geq 0$.

We then have that $(T_0(t))_{t\geq 0}$ is a nilpotent semigroup, i.e., $T(t) = 0$ for $t > r$ (see [3, II.4.31]). By [3, Corollary IV.2.5] we obtain that $\sigma(T_0(t)) = \{0\}$ for $t > 0$, thus by Inclusion (3.1) we have

$$\sigma(T_{B,0}(t)) \subset \{0\} \cup \sigma(e^{tB}). \tag{3.6}$$

Denoted by $\mathbb{D}$ the unit circle on the complex plane, we obtain that, if $(e^{tB})_{t\geq 0}$ is a hyperbolic semigroup, then $\sigma(e^{tB}) \cap \mathbb{D} = \emptyset$ for $t > 0$. Therefore, by Inclusion (3.6)

we have that $\sigma(T_{B,0}(t)) \cap \mathbb{D} = \emptyset$ for $t > 0$ yielding that the semigroup $(T_{B,0})_{t\geq 0}$ is hyperbolic as well. □

The main purpose of this section is to prove the existence of hyperbolicity of the solution semigroup $(T_{B,F,\Phi}(t))_{t\geq 0}$ under the conditions that the semigroup $(e^{tB})_{t\geq 0}$ is hyperbolic and the delay operator $\Phi$ has a sufficiently small norm. To do so, we need the following characterization of hyperbolic semigroups (see [12, Theorem 2.6.2]).

**Theorem 3.3** *Let the $(T(t))_{t\geq 0}$ be a $C_0$-semigroup on a Banach space $X$ with the generator $A$. Then the following assertions are equivalent.*

(i) *$(T(t))_{t\geq 0}$ is hyperbolic.*
(ii) *$i\mathbb{R} \subset \rho(A)$ and*

$$(C,1)\sum_{k\in\mathbb{Z}} R(i\omega + ik, A)x := \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \sum_{k=-n}^{n} R(i\omega + ik, A)x$$

*converges for all $\omega \in \mathbb{R}$ and $x \in X$.*

We note that the above theorem is taken from [12, Theorem 2.6.2], while its proof is essentially due to Greiner and Schwarz [4, Theorem 1.1 and Corollary 1.2]. A continuous version of the above theorem is proved by Kaashoek and Verduyn Lunel in [9, Theorem 4.1]. In order to apply this theorem we have to compute the resolvent $R(\lambda, G_{B,F,\Phi})$ starting from the resolvents $R(\lambda, G_{B,0})$ and $R(\lambda, G_B)$. This can be done as follows.

**Lemma 3.4** *Let the operator $(B, D(B))$ generate a hyperbolic semigroup $(e^{tB})_{t\geq 0}$. Then, if $\|\Psi\| < 1$, and $\|\Phi\|$ is sufficiently small, there exist an open strip $\Sigma$ containing the imaginary axis and a function $H_\lambda$ which is analytic and uniformly bounded on $\Sigma$ such that*

$$R(\lambda, G_{B,F,\Phi}) = H_\lambda[R(\lambda, G_{B,0}) - e_\lambda R(\lambda, B)\Psi] \quad \text{for } \lambda \in \Sigma. \tag{3.7}$$

*Proof* By [9, Theorem 4.1] and the hyperbolicity of $(e^{tB})_{t\geq 0}$ we obtain that there exist positive constants $P_1$, $\nu$ such that

$$\|R(\lambda, B)\| \leq P_1 \quad \text{for all } |\mathrm{Re}\,\lambda| < \nu.$$

We then put

$$\Sigma := \{\lambda \in \mathbb{C} : |\mathrm{Re}\,\lambda| < \nu\}$$

and

$$P := \sup_{\lambda\in\Sigma} \|R(\lambda, B)\|. \tag{3.8}$$

We first verify that for each $f \in E$ and $\lambda \in \Sigma$ the equation

$$u = e_\lambda\big[\Psi u + R(\lambda, B)\Phi u\big] - e_\lambda R(\lambda, B)\Psi f + R(\lambda, G_{B,0})f \tag{3.9}$$

has a unique solution $u \in E$.

Indeed, let $M_\lambda : E \to E$ be the linear operator defined as $M_\lambda := e_\lambda(\Psi + R(\lambda, B)\Phi)$ with $e_\lambda$ as in Theorem 2.4. For $\lambda \in \Sigma$ this operator is bounded and satisfies

$$\|M_\lambda\| \le \|\Psi\| + P\|\Phi\| < 1 \quad \text{if, in addition, } \|\Phi\| < \frac{1 - \|\Psi\|}{P}.$$

Therefore, the operator $I - M_\lambda$ is invertible, and (3.9) has a unique solution $u = (I - M_\lambda)^{-1}[R(\lambda, G_{B,0})f - e_\lambda R(\lambda, B)\Psi f]$. Putting $H_\lambda := (I - M_\lambda)^{-1}$ we obtain

$$R(\lambda, G_{B,F,\Phi}) = H_\lambda[R(\lambda, G_{B,0}) - e_\lambda R(\lambda, B)\Psi].$$

Since

$$H_\lambda = (I - M_\lambda)^{-1} = \sum_{n=0}^{\infty} M_\lambda^n \tag{3.10}$$

it follows that

$$\begin{aligned} \|H_\lambda\| &\le \sum_{n=0}^{\infty} \|M_\lambda\|^n \\ &\le \sum_{n=0}^{\infty} (\|\Psi\| + P\|\Phi\|)^n \\ &= \frac{1}{1 - \|\Psi\| - P\|\Phi\|} \quad \text{for all } \lambda \in \Sigma. \end{aligned}$$

Since $\|M_\lambda^n\| \le (\|\Psi\| + P\|\Phi\|)^n$ for all $\lambda \in \Sigma$ and the series $\sum_{n=0}^{\infty}(\|\Psi\| + P\|\Phi\|)^n$ converges for $\|\Phi\| < \frac{1-\|\Psi\|}{P}$ we obtain that, if $\|\Phi\| < \frac{1-\|\Psi\|}{P}$, then the Neumann series (3.10) converges uniformly for all $\lambda \in \Sigma$. This fact, together with the analyticity of $M_\lambda$, yields the analyticity of $H_\lambda$. □

Using the relation (3.7) and representations of resolvents we obtain the following result on the hyperbolicity of the solution semigroup $(T_{B,F,\Phi}(t))_{t\ge 0}$.

**Theorem 3.5** *Suppose that the assumptions of Theorem 2.4 are satisfied, and let the operator $(B, D(B))$ generate a hyperbolic semigroup $(e^{tB})_{t\ge 0}$. Then, if the norm of the delay operator $\Phi$ is sufficiently small, the solution semigroup $(T_{B,F,\Phi}(t))_{t\ge 0}$ is hyperbolic as well.*

*Proof* By Corollary 3.2, the semigroup $(T_{B,0}(t))_{t\ge 0}$ is hyperbolic. We first prove that, for sufficiently small $\|\Phi\|$ the sum $\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}[R(i\omega + ik, G_{B,F,\Phi})]$ is

bounded in $\mathcal{L}(E)$. In fact, by Lemma 3.4, we have

$$\begin{aligned}
&\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}[R(i\omega+ik,G_{B,F,\Phi})f](s)\\
&=\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}\left\{\left(\sum_{m=0}^{\infty}M^m_{i\omega+ik}\right)\right.\\
&\quad\left.\times[R(i\omega+ik,G_{B,0})-e_{i\omega+ik}R(i\omega+ik,B)\Psi]f\right\}(s)\\
&=\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}\left\{[R(i\omega+ik,G_{B,0})-e_{i\omega+ik}R(i\omega+ik,B)\Psi]f\right\}(s)\\
&\quad+\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}e^{(i\omega+ik)s}(\Psi+R(i\omega+ik,B)\Phi)\{[R(i\omega+ik,G_{B,0})\\
&\quad-e_{i\omega+ik}R(i\omega+ik,B)\Psi]f\}(s)+\cdots
\end{aligned}\tag{3.11}$$

for $s\in[-r,0]$.

Note that the semigroup $(T_{B,0}(t))_{t\geq 0}$ is hyperbolic, hence $e^{-2\pi i\omega}\in\rho(T_{B,0}(2\pi))$ $\forall\omega\in\mathbb{R}$. Using the formula (see [3, Lemma II.1.9])

$$R(\lambda,G_{B,0})(1-e^{-\lambda t}T_{B,0}(t))=\int_0^t e^{-\lambda s}T_{B,0}(s)ds,\quad \lambda\in\rho(G_{B,0})$$

we obtain

$$\begin{aligned}
R(i\omega+ik,G_{B,0})&=\int_0^{2\pi}e^{-(i\omega+ik)t}T_{B,0}(t)(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1}dt,\\
R(i\omega+ik,B)&=\int_0^{2\pi}e^{-(i\omega+ik)t}e^{tB}(1-e^{2\pi B})^{-1}dt.
\end{aligned}$$

The first term of (3.11) can now be computed as

$$\begin{aligned}
&\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}[R(i\omega+ik,G_{B,0})-e_{i\omega+ik}R(i\omega+ik,B)\Psi]f\\
&=\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}\int_0^{2\pi}e^{-(i\omega+ik)t}[T_{B,0}(t)(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1}\\
&\quad-e_{i\omega+ik}e^{tB}(1-e^{2\pi B})^{-1}\Psi]f\,dt\\
&=\int_0^{2\pi}\left[\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}e^{-ikt}\right]e^{-i\omega t}[T_{B,0}(t)(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1}
\end{aligned}$$

$$
\begin{aligned}
&- e_{\omega+ik}e^{tB}(1-e^{2\pi B})^{-1}\Psi]f\,dt \\
&= \int_0^{2\pi} \sigma_N(t)e^{-i\omega t}[T_{B,0}(t)(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1} \\
&\quad - e_{i\omega+ik}e^{tB}(1-e^{2\pi B})^{-1}\Psi]f\,dt.
\end{aligned}
$$

Here, $\sigma_N(t) = \frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n} e^{-ikt}$. Since

$$
\sigma_N(t) = \frac{1-\cos(Nt)}{N(1-\cos t)} \geq 0 \quad \text{and} \quad \int_0^{2\pi} \sigma_N(t)dt = 2\pi \tag{3.12}
$$

(see [4, Theorem 1.1]), the norm of the first term in (3.11) can be estimated by

$$
\left\| \frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n} \big[R(i\omega+ik, G_{B,0}) - e_{i\omega+ik}R(i\omega+ik, B)\Psi\big]f \right\| \leq C_1\|f\| \tag{3.13}
$$

with

$$
\begin{aligned}
C_1 := 2\pi \sup_{0\leq\omega\leq 1} &\{\|(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1}\| + \|(1-e^{2\pi B})^{-1}\|\|\Psi\|\} \\
&\times \sup_{0\leq t\leq 2\pi} \{\|T_{B,0}(t)\| + \|e^{tB}\|\}.
\end{aligned}
$$

We now compute the second term of (3.11). For $s \in [-r, 0]$ we have

$$
\begin{aligned}
&\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n} M_{(i\omega+ik)}[(R(i\omega+ik, G_{B,0}) - e_{i\omega+ik}R(i\omega+ik, B)\Psi)f](s) \\
&= \frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n} e^{(i\omega+ik)s}(\Psi + R(i\omega+ik, B)\Phi)[R(i\omega+ik, G_{B,0}) \\
&\quad - e_{i\omega+ik}R(i\omega+ik, B)\Psi]f \\
&= \frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n} e^{(i\omega+ik)s}\left(\Psi + \int_0^{2\pi} e^{-(i\omega+ik)\tau}e^{\tau B}(1-e^{2\pi B})^{-1}d\tau\,\Phi\right) \\
&\int_0^{2\pi} e^{-(i\omega+ik)t}\left[T_{B,0}(t)(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1} - e_{i\omega+ik}e^{tB}(1-e^{2\pi B})^{-1}\Psi\right]f\,dt \\
&= \int_0^{2\pi}\left[\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n} e^{-ik(t-s)}\right]e^{-i\omega(t-s)}\Psi\left[T_{B,0}(t)(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1}\right. \\
&\quad \left. - e_{i\omega+ik}e^{tB}(1-e^{2\pi B})^{-1}\Psi\right]f\,dt \\
&\quad + \int_0^{2\pi}\int_0^{2\pi}\left[\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n} e^{-ik(t+\tau-s)}\right]e^{-i\omega(t+\tau-s)}e^{\tau B}(1-e^{2\pi B})^{-1}\Phi
\end{aligned}
$$

$$\times [T_{B,0}(t)(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1} - e_{i\omega+ik}e^{tB}(1-e^{2\pi B})^{-1}\Psi]fd\tau dt$$
$$= \int_0^{2\pi} \sigma_N(t-s)e^{-i\omega(t-s)}\Psi[T_{B,0}(t)(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1}$$
$$- e_{i\omega+ik}e^{tB}(1-e^{2\pi B})^{-1}\Psi]fdt$$
$$+ \int_0^{2\pi}\int_0^{2\pi} \sigma_N(t+\tau-s)e^{-i\omega(t+\tau-s)}e^{\tau B}(1-e^{2\pi B})^{-1}\Phi$$
$$\times [T_{B,0}(t)(1-e^{-2\pi i\omega}T_{B,0}(2\pi))^{-1} - e_{i\omega+ik}e^{tB}(1-e^{2\pi B})^{-1}\Psi]fd\tau dt.$$

Therefore, using (3.12), the norm of the second term of (3.11) can be estimated by

$$C_1(\|\Psi\| + C_2\|\Phi\|)\|f\|$$

with $C_2 := 2\pi\|(1-e^{2\pi B})^{-1}\| \sup_{0\le t\le 2\pi}\{\|e^{tB}\|\}$ and $C_1$ as in (3.13), respectively. By induction, the norm of the $n$th term of (3.11) is estimated by

$$C_1(\|\Psi\| + C_2\|\Phi\|)^n\|f\|.$$

Moreover, the series $\sum_{n=0}^{\infty} C_1(\|\Psi\| + C_2\|\Phi\|)^n\|f\|$ converges if $\|\Phi\| < \frac{1-\|\Psi\|}{C_2}$.

Hence, for these $\|\Phi\|$ the sum $\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n} R(i\omega+ik, G_{B,F,\Phi})$ is bounded in $\mathcal{L}(E)$.

We now prove the convergence of $\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}[R(i\omega+ik, G_{B,F,\Phi})f]$ for $\omega\in\mathbb{R}$, $f\in E$. This can be done by using the idea from [4, Theorem 1.1]. By [13, III.4.5], it is sufficient to show the convergence on a dense subset. From the fact that $i\mathbb{R}\subset\rho(G_{B,F,\Phi})$ and by the spectral mapping theorem for the residual spectrum (see [3, Theorem IV.3.7]) we obtain that $e^{-2\pi i\omega}$ does not belong to the residual spectrum $R\sigma(T_{B,F,\Phi})$. This implies that $(1-e^{-2\pi i\omega}T_{B,F,\Phi}(2\pi))E$ is a dense subset of $E$. Let $f = (1-e^{-2\pi i\omega}T_{B,F,\Phi}(2\pi))g$. Then

$$\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n} R(i\omega+ik, G_{B,F,\Phi})(1-e^{-2\pi i\omega}T_{B,F,\Phi}(2\pi))g$$
$$= \frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=-n}^{n}\int_0^{2\pi} e^{-(i\omega+ik)s}T_{B,F,\Phi}(s)gds. \tag{3.14}$$

Now, $e^{-i\omega}T_{B,F,\Phi}(\cdot)g$ is a continuous function with Fourier coefficients

$$Q_k = \frac{1}{2\pi}\int_0^{2\pi} e^{-(i\omega+ik)s}T_{B,F,\Phi}(s)gds.$$

Therefore, by Fejer's Theorem [10, Theorem I.3.1], the sum in (3.14) converges as $N\to\infty$. The assertion of the theorem now follows from Theorem 3.3. □

## 4 An example

Consider the neutral partial differential equation

$$\begin{aligned}\frac{\partial w(x,t)}{\partial t} - k\frac{\partial w(x,t-1)}{\partial t} &= \frac{\partial^2 w(x,t)}{\partial x^2} - k\frac{\partial^2 w(x,t-1)}{\partial x^2} + \alpha w(x,t)\\ &\quad + \int_{-1}^{0} \psi(s) w(x,t+s)ds \qquad \text{for } 0 \le x \le \pi,\ t \ge 0,\\ w(0,t) = w(\pi,t) &= 0, \quad t \ge 0,\\ w(x,s) &= \varphi(x,s) \quad \text{for all } x \in [0,\pi],\ s \in [-1,0],\end{aligned} \tag{4.1}$$

where $k$ and $\alpha$ are real constants with $|k| < 1$, $\alpha > 1$ and $\alpha \neq n^2\ \forall n \in \mathbb{N}$; the functions $\psi$ and $\varphi$ are given such that $\psi \in L_1[-1,0]$ and $\varphi$ is continuous.

We choose the Hilbert space $X := L_2[0,\pi]$ and let $B : X \to X$ be defined by

$$B(f) = f'' + \alpha f$$

with the domain $D(B) = H_0^2[0,\pi] := \{f \in W^{2,2}[0,\pi] : f(0) = f(\pi) = 0\}$.

Also define the difference and delay operators $F$ and $\Phi$ as

$$F : C([-1,0],X) \to X, \quad F(f) := f(0) - kf(-1)$$

$$\Phi : C([-1,0],X) \to X, \quad \Phi(f) := \int_{-1}^{0} \psi(s) f(s) ds.$$

Clearly, $F$ and $\Phi$ are bounded linear operators. Moreover, $\|\Phi\| \le \|\psi\|_{L_1}$.

Equation (4.1) can now be rewritten as

$$\begin{cases} \frac{\partial}{\partial t} F u_t = BFu_t + \Phi u_t, & \text{for } t \ge 0,\\ u_0(s) = \varphi(\cdot, s), & \text{for } s \in [-1,0] \end{cases} \tag{4.2}$$

for $u(t) = w(\cdot,t)$.

It can be seen (see [3]) that $B$ is the generator of an analytic semigroup $(e^{tB})_{t\ge 0}$.

Since $\sigma(B) = \{-1+\alpha, -4+\alpha, \ldots, -n^2+\alpha, \ldots\}$ and $\alpha \neq n^2\ \forall n \in \mathbb{N}$, it follows that $\sigma(B) \cap i\mathbb{R} = \emptyset$. Therefore, applying the spectral mapping theorem for analytic semigroups we obtain that the semigroup $(e^{tB})_{t\ge 0}$ is hyperbolic.

Theorem 3.4 now yields that, if $\|\psi\|_{L_1} < \frac{1-|k|}{2\pi \|(1-e^{2\pi B})^{-1}\| \sup_{0\le t\le 2\pi}\{\|e^{tB}\|\}}$, then the solution semigroup $(T_{B,F,\Phi}(t))_{t\ge 0}$ for (4.2) is also hyperbolic.

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